1.61 Lines and Planes

Worked Solutions

1 a)

The normal vector of the plane $x = 0$ is $\mathbf{n} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, and the line has direction vector $\mathbf{d} = \begin{pmatrix} 2 \\ 4 \\ -9 \end{pmatrix}$.

Thus

$$\cos\theta = \frac{\mathbf{n}\cdot\mathbf{d}}{|\mathbf{n}||\mathbf{d}|} = \frac{2}{\sqrt{94}}, \qquad \theta = \arccos\left(\frac{2}{\sqrt{94}}\right) = 78.095\ldots^\circ.$$

The angle between the line and plane is $90^\circ - 78.095\ldots^\circ = 11.9^\circ$ (1 d.p.).

1 b)

The normal vector of the plane $y = 0$ is $\mathbf{n} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$. Hence

$$\cos\theta = \frac{3}{\sqrt{94}}, \qquad \theta = \arccos\left(\frac{3}{\sqrt{94}}\right) = 71.975\ldots^\circ.$$

The angle between the line and plane is $90^\circ - 71.975\ldots^\circ = 18.0^\circ$ (1 d.p.).

1 c)

The normal vector of the plane $z = 0$ is $\mathbf{n} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$. Thus

$$\cos\theta = -\frac{9}{\sqrt{94}}, \qquad \theta = \arccos\left(-\frac{9}{\sqrt{94}}\right) = 158.168\ldots^\circ.$$

The acute angle between the line and the normal is $180^\circ - 158.168\ldots^\circ = 21.831\ldots^\circ$, so the angle between the line and plane is $90^\circ - 21.831\ldots^\circ = 68.2^\circ$ (1 d.p.).

2)

The plane has normal vector $\mathbf{n} = \begin{pmatrix} 4 \\ 1 \\ -2 \end{pmatrix}$ and the line direction vector is $\mathbf{d} = \begin{pmatrix} 2 \\ 4 \\ -3 \end{pmatrix}$. Therefore

$$\cos\theta = \frac{\mathbf{n}\cdot\mathbf{d}}{|\mathbf{n}||\mathbf{d}|} = \frac{18}{\sqrt{21}\sqrt{29}} = \frac{6\sqrt{609}}{203}.$$

So $\theta = \arccos\left(\frac{6\sqrt{609}}{203}\right) = 43.164\ldots^\circ$, and the acute angle between the line and plane is $90^\circ - 43.164\ldots^\circ = 46.8^\circ$ (1 d.p.).

3)

A normal vector to the plane is $\mathbf{n} = \begin{pmatrix} 6 \\ 15 \\ 10 \end{pmatrix}$ and the line direction is $\mathbf{d} = \begin{pmatrix} 5 \\ -8 \\ 9 \end{pmatrix}$. Their dot product is $30 - 120 + 90 = 0$, so the line is parallel to the plane. The point $(3, 7, -3)$ on the line does not lie on the plane, since $6(3) + 15(7) + 10(-3) - 90 = 3 \neq 0$. Thus the line is parallel to, but not contained in, the plane.

4)

A normal vector to the plane is $\mathbf{n} = \begin{pmatrix} 3 \\ 12 \\ -6 \end{pmatrix}$ and the line direction is $\mathbf{d} = \begin{pmatrix} 6 \\ -4 \\ -5 \end{pmatrix}$. Since $\mathbf{n} \cdot \mathbf{d} = 18 - 48 + 30 = 0$, the line is parallel to the plane. Its point $(2, \frac{1}{4}, \frac{11}{3})$ lies on the plane because

$$3(2) + 12\left(\frac{1}{4}\right) - 6\left(\frac{11}{3}\right) + 13 = 6 + 3 - 22 + 13 = 0.$$

Therefore the line is contained in the plane.

5)

The plane has normal vector $\mathbf{n} = \begin{pmatrix} 5 \\ 3 \\ 2 \end{pmatrix}$ and the line direction is $\mathbf{d} = \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix}$. Since $\mathbf{n} \cdot \mathbf{d} = -10 + 3 + 8 = 1 \neq 0$, the line is not parallel to the plane and so intersects it at one point.

6 a)

The plane has normal vector $\mathbf{n} = \begin{pmatrix} 1 \\ 1 \\ 10 \end{pmatrix}$. A direction vector for the pipe is

$$\mathbf{d} = \begin{pmatrix} 1 \\ 1 \\ -0.5 \end{pmatrix} - \begin{pmatrix} 0.2 \\ 0.4 \\ -0.6 \end{pmatrix} = \begin{pmatrix} 0.8 \\ 0.6 \\ 0.1 \end{pmatrix},$$

which can be scaled to $\begin{pmatrix} 8 \\ 6 \\ 1 \end{pmatrix}$. Hence

$$\cos\theta = \frac{24}{\sqrt{102}\sqrt{101}} = \frac{4\sqrt{10302}}{1717}, \qquad \theta = 76.322\ldots^\circ.$$

The angle between the line and plane is $90^\circ - 76.322\ldots^\circ = 13.7^\circ$ (1 d.p.).

6 b)

A direction vector of the pipe is $\mathbf{d} = \begin{pmatrix} 8 \\ 6 \\ 1 \end{pmatrix}$. Let A be a point on the pipe with position vector $\overrightarrow{OA} = \begin{pmatrix} 1 + 8\lambda \\ 1 + 6\lambda \\ -0.5 + \lambda \end{pmatrix}$. Then

$$\overrightarrow{PA} = \begin{pmatrix} 0.2 + 8\lambda \\ -91.1 + 6\lambda \\ 8.69 + \lambda \end{pmatrix}.$$

At the closest point, $\overrightarrow{PA} \cdot \mathbf{d} = 0$, so

$$8(0.2 + 8\lambda) + 6(-91.1 + 6\lambda) + (8.69 + \lambda) = 0 \quad \Rightarrow \quad 101\lambda - 536.31 = 0,$$

giving $\lambda = 5.31$. Thus $\overrightarrow{PA} = \begin{pmatrix} 42.68 \\ -59.24 \\ 14 \end{pmatrix}$ and

$$|PA| = \sqrt{42.68^2 + (-59.24)^2 + 14^2} = 74.3 \text{ m (1 d.p.)}.$$

Alternatively, using the vector product, with $\mathbf{p} = \begin{pmatrix} 0.8 \\ 92.1 \\ -9.19 \end{pmatrix}$, $\mathbf{a} = \begin{pmatrix} 1 \\ 1 \\ -0.5 \end{pmatrix}$ and $\mathbf{d} = \begin{pmatrix} 8 \\ 6 \\ 1 \end{pmatrix}$,

$$h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left| \begin{pmatrix} 8 \\ 6 \\ 1 \end{pmatrix} \times \begin{pmatrix} -0.2 \\ 91.1 \\ -8.69 \end{pmatrix} \right|}{\sqrt{101}} = 74.3 \text{ m (1 d.p.).}$$

7 a)

The line has $x = 2 + 5\lambda$, $y = -1 + \lambda$, $z = 3 - 2\lambda$. For $x = 0$, $\lambda = -\frac{2}{5}$, giving $(0, -\frac{7}{5}, \frac{19}{5})$.

7 b)

For $y = 0$, $\lambda = 1$, giving $(7, 0, 1)$.

7 c)

For $z = 0$, $\lambda = \frac{3}{2}$, giving $(\frac{19}{2}, \frac{1}{2}, 0)$.

8)

The line has $x = 3$, $y = 1 + 2\lambda$ and $z = 1 + 5\lambda$. Substituting into $4x + y - z = 9$ gives $12 + (1 + 2\lambda) - (1 + 5\lambda) = 9$, so $\lambda = 1$. The intersection point is $(3, 3, 6)$.

9)

The line through A perpendicular to the plane has equation $\mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$. Its intersection with the plane satisfies $(3 + \lambda) + (2 + \lambda) - (4 - \lambda) = 0$, so $\lambda = -\frac{1}{3}$ and the foot of the perpendicular is $P = (\frac{8}{3}, \frac{5}{3}, \frac{13}{3})$. Then

$$\overrightarrow{AP} = \begin{pmatrix} -\frac{1}{3} \\ -\frac{1}{3} \\ \frac{1}{3} \end{pmatrix}, \qquad \overrightarrow{AB} = 2\overrightarrow{AP},$$

so the reflected image is $B = (\frac{7}{3}, \frac{4}{3}, \frac{14}{3})$.

10)

A line through P perpendicular to the plane has equation $\mathbf{r} = \begin{pmatrix} 5 \\ -2 \\ \frac{1}{2} \end{pmatrix} + \lambda \begin{pmatrix} 10 \\ -1 \\ -2 \end{pmatrix}$ (a scaled normal vector). Substitution into $10x - y - 2z = -54$ gives $\lambda = -1$. The foot of the perpendicular is $A = (-5, -1, \frac{5}{2})$, so the reflected image is

$$P' = P + 2(A - P) = (-15, 0, \frac{9}{2}).$$

11)

For l_1, $x = 2 + \lambda$, $y = 8 + 4\lambda$, $z = 3$. Substitution into $9x - 20y + 2z = 6$ gives $-71\lambda - 142 = 0$, so $\lambda = -2$ and $P = (0, 0, 3)$. For l_2, $x = -1 + 2\mu$, $y = 0$, $z = \mu$. Substitution gives $20\mu - 15 = 0$, so $\mu = \frac{3}{4}$ and $Q = (\frac{1}{2}, 0, \frac{3}{4})$. Hence

$$|PQ| = \sqrt{\left(\frac{1}{2}\right)^2 + 0^2 + \left(-\frac{9}{4}\right)^2} = \frac{\sqrt{85}}{4}.$$

12 a)

If the line lies in the plane, its direction vector is perpendicular to the normal vector, so

$$\begin{pmatrix} b \\ 4 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = 2b + 5 = 0,$$

which gives $b = -\frac{5}{2}$. Its point $(2, a, 3)$ must also lie on the plane, so $4 + a + 3 = 5$ and $a = -2$. Therefore $a = -2$ and $b = -\frac{5}{2}$.

12 b)

The line's direction vector is perpendicular to the plane normal when $2b + 5 = 0$, i.e. $b = -\frac{5}{2}$. For any other value of b, it intersects the plane at exactly one point, independently of a. Thus a is arbitrary and $b \neq -\frac{5}{2}$.

13 a)

The point P lies on the line when its coordinates satisfy $-3 = -\lambda$, $7 = 1 + 2\lambda$ and $5 = 2 + \lambda$. Each gives $\lambda = 3$, so P lies on l.

13 b)

The plane normal is parallel to the line direction, so its equation is $-x + 2y + z = d$. Substituting $P = (-3, 7, 5)$ gives $d = 3 + 14 + 5 = 22$. Hence $\Pi : -x + 2y + z = 22$.

13 c)

Since R lies on the line, $R = (-\lambda, 1 + 2\lambda, 2 + \lambda)$. Using $Q = (-3, 7 - \frac{1}{\sqrt{5}}, 5 + \frac{2}{\sqrt{5}})$ and $QR = 5$ gives

$$(-\lambda + 3)^2 + \left(2\lambda - 6 + \frac{1}{\sqrt{5}}\right)^2 + \left(\lambda - 3 - \frac{2}{\sqrt{5}}\right)^2 = 25.$$

This simplifies to $6\lambda^2 - 36\lambda + 30 = 0$, or $(\lambda - 1)(\lambda - 5) = 0$. Thus the possible coordinates are $(-1, 3, 3)$ and $(-5, 11, 7)$.

14 a)

$\overrightarrow{AB} = B - A = \begin{pmatrix} 1 \\ 1 \\ -5 \end{pmatrix}$, so l_1 is

$$\mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -5 \end{pmatrix}.$$

14 b)

$\overrightarrow{AC} = C - A = \begin{pmatrix} -2 \\ 4 \\ -2 \end{pmatrix}$, so

$$l_2 : \ \mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 4 \\ -2 \end{pmatrix},$$

or, using a scaled direction vector, $\mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$.

14 c)

The direction vectors of l_1 and l_2 are perpendicular to $\mathbf{n} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$ because

$$\begin{pmatrix} 1 \\ 1 \\ -5 \end{pmatrix} \cdot \mathbf{n} = 3 + 2 - 5 = 0, \qquad \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \cdot \mathbf{n} = 3 - 4 + 1 = 0.$$

14 d)

The plane has equation $3x + 2y + z + d = 0$. Substituting $A = (3, -2, 3)$ gives $9 - 4 + 3 + d = 0$, hence $d = -8$. Therefore $\Pi : 3x + 2y + z - 8 = 0$.

14 e)

$\overrightarrow{DE} = E - D = \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$, so l_3 has equation $\mathbf{r} = \begin{pmatrix} 8 \\ 5 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$. Substituting into the plane gives

$$3(8 - 2\lambda) + 2(5 - \lambda) + (9 + \lambda) - 8 = 0 \quad \Rightarrow \quad \lambda = 5.$$

Thus $P = (-2, 0, 14)$.

14 f)

The normal is $\mathbf{n} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$ and line direction is $\mathbf{d} = \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$. Then $\cos\theta = -\frac{7}{\sqrt{14}\sqrt{6}}$, so $\theta = 139.797\ldots^\circ$.
The acute angle between the line and plane is $90^\circ - (180^\circ - 139.797\ldots^\circ) = 49.8^\circ$ (1 d.p.).

14 g)

A unit vector in the direction of $\mathbf{d}$ is $\frac{1}{\sqrt{6}} \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$. Hence

$$F = P \pm \frac{1}{\sqrt{6}} \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix},$$

so the possible coordinates are

$$\left(-2 - \frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, 14 + \frac{1}{\sqrt{6}}\right) \quad \text{or} \quad \left(-2 + \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, 14 - \frac{1}{\sqrt{6}}\right).$$

15 a)

The direction vector of the line has dot product $3(1) - 9(4) + 2(3) = -27 \neq 0$ with the plane normal, so the line intersects the plane once. Substituting $\mathbf{r} = \begin{pmatrix} 2 \\ 5 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}$ into $3x - 9y + 2z = 18$ gives $-27\lambda - 20 = 0$, so $\lambda = -\frac{20}{27}$. The intersection point is $\left(\frac{34}{27}, -\frac{26}{27}, \frac{25}{9}\right)$.

15 b)

The line direction has dot product $3(7) - 9(13) + 2(48) = 0$ with the plane normal, so the line is parallel to the plane. Its point $(6, -3, 2)$ does not lie in the plane because $3(6) - 9(-3) + 2(2) = 49 \neq 18$. Thus it is parallel to, but not contained in, the plane.

15 c)

The line has point $(6, 4, 18)$ and direction $(2, 4, 15)$. Its direction is perpendicular to the plane normal since $3(2) - 9(4) + 2(15) = 0$. Also $3(6) - 9(4) + 2(18) = 18$, so the line lies in the plane.

15 d)

Write the line as $\mathbf{r} = \begin{pmatrix} -8 \\ 3 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 6 \end{pmatrix}$. The direction vector has dot product $3 + 9 + 12 = 24 \neq 0$ with the plane normal, so it intersects once. Substitution into $3x - 9y + 2z = 18$ gives $24\lambda - 71 = 0$, so $\lambda = \frac{71}{24}$. The intersection point is $\left(-\frac{121}{24}, \frac{1}{24}, \frac{67}{4}\right)$.

16 a)

The lines can be written as

$$l_1 : \ \mathbf{r} = \begin{pmatrix} -5 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}, \qquad l_2 : \ \mathbf{r} = \begin{pmatrix} 5 \\ -3 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}.$$

Equating coordinates gives $3\lambda + \mu = 10$, $\lambda - \mu = -4$ and $-\lambda - 2\mu = 3$. The first two give $\lambda = \frac{3}{2}$, $\mu = \frac{11}{2}$, which do not satisfy the third equation. The lines do not intersect. Their direction vectors are not parallel, so they are skew.

16 b)

A vector perpendicular to both directions is their cross product:

$$\begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} \times \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ -5 \\ 4 \end{pmatrix}.$$

16 c)

A plane parallel to both lines has normal vector $(3, -5, 4)$. To contain the point $(5, -3, 5)$ on l_2, it must satisfy $3x - 5y + 4z = 50$. Thus $\Pi_1 : 3x - 5y + 4z = 50$.

16 d)

Using the point $(-5, 1, 2)$ on l_1 gives $3(-5) - 5(1) + 4(2) = -12$. Thus $\Pi_2 : 3x - 5y + 4z = -12$.

16 e)

Substituting l_3 into Π_1 gives $8\alpha + 5 = 50$, so $\alpha = \frac{45}{8}$ and $P = (-\frac{119}{8}, \frac{175}{8}, 51)$. Substituting into Π_2 gives $8\alpha + 5 = -12$, so $\alpha = -\frac{17}{8}$ and $Q = (\frac{67}{8}, -\frac{11}{8}, -11)$. Therefore

$$\overrightarrow{PQ} = Q - P = \begin{pmatrix} \frac{93}{4} \\ -\frac{93}{4} \\ -62 \end{pmatrix} = \frac{31}{4} \begin{pmatrix} 3 \\ -3 \\ -8 \end{pmatrix}, \qquad PQ = \frac{31\sqrt{82}}{4}.$$

17)

The line can be written as $\mathbf{r} = \begin{pmatrix} -4 \\ 4 \\ 2 \end{pmatrix} + \alpha \begin{pmatrix} \frac{1}{2} \\ -3 \\ 0 \end{pmatrix}$. A normal vector to the plane is

$$\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} \times \begin{pmatrix} 7 \\ 3 \\ 5 \end{pmatrix} = \begin{pmatrix} -11 \\ 4 \\ 13 \end{pmatrix}.$$

Using the point $(1, 1, 4)$, the plane equation is $-11x + 4y + 13z - 45 = 0$. Substituting the line gives $\frac{35}{2}\alpha - 41 = 0$, so $\alpha = \frac{82}{35}$. The point of intersection is $\left(-\frac{99}{35}, -\frac{106}{35}, 2\right)$.

18 a)

Take $\mathbf{b} = (50, 120, 235) - (60, 70, 320) = (-10, 50, -85)$ and $\mathbf{c} = (130, 100, 155) - (50, 120, 235) = (80, -20, -80)$. The plane is

$$\mathbf{r} = \begin{pmatrix} 60 \\ 70 \\ 320 \end{pmatrix} + \lambda \begin{pmatrix} -10 \\ 50 \\ -85 \end{pmatrix} + \mu \begin{pmatrix} 80 \\ -20 \\ -80 \end{pmatrix}.$$

18 b)

The vector $(3, 4, 2)$ is perpendicular to the plane because

$$(3, 4, 2) \cdot (-10, 50, -85) = -30 + 200 - 170 = 0, \qquad (3, 4, 2) \cdot (80, -20, -80) = 240 - 80 - 160 = 0.$$

18 c)

The plane has equation $3x + 4y + 2z = d$. Substituting $(60, 70, 320)$ gives $d = 1100$, so $3x + 4y + 2z = 1100$.

18 d)

The perpendicular line through $(20, 65, 100)$ is $\mathbf{r} = (20, 65, 100) + \alpha(3, 4, 2)$. Substituting into the plane gives

$$3(20 + 3\alpha) + 4(65 + 4\alpha) + 2(100 + 2\alpha) = 1100 \quad \Rightarrow \quad \alpha = 20.$$

The closest point on the rock face is $(80, 130, 140)$. The tunnel length is

$$\sqrt{(80 - 20)^2 + (130 - 65)^2 + (140 - 100)^2} = 20\sqrt{29} \text{ m}.$$

Alternatively, with normal $(3, 4, 2)$, plane $3x + 4y + 2z - 1100 = 0$, and point $(20, 65, 100)$, the point-to-plane distance is

$$\frac{|3(20) + 4(65) + 2(100) - 1100|}{\sqrt{3^2 + 4^2 + 2^2}} = \frac{580}{\sqrt{29}} = 20\sqrt{29} \text{ m}.$$

19 a)

The plane normals have dot product $(4, -1, -1) \cdot (3, 2, 2) = 12 - 2 - 2 = 8 \neq 0$, so Π_1 and Π_2 are not perpendicular.

19 b)

Set $z = 0$. Solving $4x - y = 3$ and $3x + 2y = 9$ gives $x = \frac{15}{11}$ and $y = \frac{27}{11}$. Thus $(\frac{15}{11}, \frac{27}{11}, 0)$ lies on both planes.

19 c)

A direction vector of the line of intersection is the cross product of the plane normals:

$$(4, -1, -1) \times (3, 2, 2) = (0, -11, 11),$$

so $(0, -1, 1)$ is parallel to both planes.

19 d)

The plane perpendicular to both has normal parallel to $(0, -1, 1)$, so it has equation $-y + z = d$. Substituting $(5, 3, 1)$ gives $d = -2$. Hence $\Pi_3 : -y + z = -2$.

20 a)

Substitution of the line into the plane gives

$$5(4 + 2\lambda) + 8(3 + 6\lambda) + 9(-5 + 3\lambda) = 16 \quad \Rightarrow \quad 85\lambda - 17 = 0,$$

so the line intersects the plane at $(\frac{22}{5}, \frac{21}{5}, -\frac{22}{5})$. The line direction has length 7, so a vector of length 14 along it is $2(2, 6, 3) = (4, 12, 6)$. Therefore

$$P = \left(\frac{42}{5}, \frac{81}{5}, \frac{8}{5}\right), \qquad Q = \left(\frac{2}{5}, -\frac{39}{5}, -\frac{52}{5}\right),$$

up to their order.

20 b)

Let B be a point on the line with position vector $(4+2\lambda, 3+6\lambda, -5+3\lambda)$. For AB to be perpendicular to the line direction $(2, 6, 3)$,

$$(-7 + 2\lambda, 9 + 6\lambda, -6 + 3\lambda) \cdot (2, 6, 3) = 0 \quad \Rightarrow \quad 49\lambda + 22 = 0,$$

so $\lambda = -\frac{22}{49}$. The shortest distance from A to the line is $\frac{15\sqrt{34}}{7}$. Since $PQ = 28$, the triangle area is

$$\frac{1}{2} \cdot 28 \cdot \frac{15\sqrt{34}}{7} = 30\sqrt{34} \text{ units}^2.$$

Alternatively, using the vector product, with $\mathbf{d} = (2, 6, 3)$ and $\mathbf{p} - \mathbf{a} = (7, -9, 6)$,

$$h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{|(63, 9, -60)|}{7} = \frac{15\sqrt{34}}{7},$$

which gives the same area.

21)

The line l meets the plane when $(2 + \lambda) - (3 - 7\lambda) + (-8 + 6\lambda) = 6$, giving $\lambda = \frac{15}{14}$. Thus the intersection is $(\frac{43}{14}, -\frac{9}{2}, -\frac{11}{7})$. Reflect the point $(2, 3, -8)$ in the plane using the perpendicular direction $(1, -1, 1)$. The perpendicular line meets the plane at $(7, -2, -3)$, so the reflected point is

$$(2, 3, -8) + 2(5, -5, 5) = (12, -7, 2).$$

The reflected line passes through $(12, -7, 2)$ and $(\frac{43}{14}, -\frac{9}{2}, -\frac{11}{7})$. A direction vector is proportional to $(25, -7, 10)$, so an equation is

$$\mathbf{r} = \begin{pmatrix} 12 \\ -7 \\ 2 \end{pmatrix} + \alpha \begin{pmatrix} 25 \\ -7 \\ 10 \end{pmatrix}.$$

22)

The direction vector $(16, -50, 23)$ is perpendicular to both plane normals:

$$(16,-50,23)\cdot(6,1,-2) = 96-50-46 = 0, \qquad (16,-50,23)\cdot(5,-3,-10) = 80+150-230 = 0.$$

The point $(22, -64, 30)$ lies on both planes because $6(22) - 64 - 2(30) = 8$ and $5(22) - 3(-64) - 10(30) = 2$. Therefore the line lies on both planes.

23)

A direction vector for the intersection is

$$(2,-1,1)\times(3,1,-1) = (0,5,5),$$

so we can use $(0, 1, 1)$. Setting $y = 1$ in the two plane equations gives $2x + z = 6$ and $3x - z = 1$, so $x = \frac{7}{5}$ and $z = \frac{16}{5}$. The line is

$$\mathbf{r} = \begin{pmatrix} \frac{7}{5} \\ 1 \\ \frac{16}{5} \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

24)

A direction vector for the intersection is

$$(0,0,1)\times(5,1,3) = (-1,5,0).$$

Taking $x = 1$ and $z = 0$ gives $y = 3$, so the line is

$$\mathbf{r} = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 5 \\ 0 \end{pmatrix}.$$

25 a)

The line of intersection of the wall and hutch planes has direction

$$(1,0,0)\times(1,0,-4) = (0,4,0),$$

which can be scaled to $(0, 1, 0)$. On the wall $x = 0$ and on the hutch plane $x - 4z + 20 = 0$, so $z = 5$. Taking $y = 1$ gives a point on the line, and thus

$$\mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

25 b)

The angle between the two planes is the acute angle between their normal vectors $(1, 0, 0)$ and $(1, 0, -4)$. Hence

$$\cos\theta = \frac{1}{\sqrt{17}}, \qquad \theta = \arccos\left(\frac{1}{\sqrt{17}}\right) = 75.963\ldots^\circ = 76.0^\circ \text{ (3 s.f.)}.$$